

Additional file 1

Additional file 1, Table A: default hyperparameters for all 19 model configurations in the study. Every value is read from `models/model_defaults.py` at spec version 1.7.0, spec hash 26163cc378cd, which is the file both the QM9 pipeline and the assay pipeline load their settings from. Every QM9 results row carries the spec version and the spec hash as two of its columns, so a QM9 table can be traced back to the settings that produced it; the three assay datasets are written by a second pipeline whose results files do not carry those two columns. The configuration list is the model roster the job generator submits. One setting is used for every representation and for every dataset, except where Table B gives a tuned replacement. The support vector machine uses an RBF kernel on every representation, so no result in this study carries a kernel that changes with the representation. Two values are not held in the shared spec and name their source file in the third column.

Configuration	Hyperparameter	Default value
<i>Shared by every neural network below</i>		
Training, all networks	optimizer	adam
	lr	0.001
	weight_decay	0.0
	batch_size	32
	epochs	100
		cap; early stopping usually reaches it first
	patience	10
		epochs without a strictly better mean validation loss
	restore_best_weights	True
		the reported fit is the best epoch, not the last
	val_loss_reduction	mean
	improvement_tolerance	0.0
	mc_passes	100
		stochastic forward passes at inference
	standardise_targets	True
		fitted on training labels alone
<i>Tree and boosting models</i>		
RF	n_estimators	100
rf	max_depth	None
		no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
XGBoost	n_estimators	100
xgboost	max_depth	6
	learning_rate	0.1

Configuration	Hyperparameter	Default value
LightGBM lgb	subsample	1.0
	colsample_bytree	1.0
	colsample_bylevel	1.0
	min_child_weight	1
	gamma	0.0
	reg_alpha	0.0
	reg_lambda	1.0
	n_estimators	100
	learning_rate	0.1
	num_leaves	31
NGBoost ngboost	max_depth	-1
		no limit
	subsample	1.0
	colsample_bytree	1.0
	min_child_samples	20
	reg_alpha	0.0
	reg_lambda	0.0
	n_estimators	500
	learning_rate	0.01
	natural_gradient	True
QRF qrf	dist	Normal
		ngboost.distns.Normal
	score	MLE
		ngboost.scores.MLE, the log scoring rule
	minibatch_frac	1.0
	col_sample	1.0
	base_max_depth	3
		depth of one boosting stage
	base_criterion	friedman_mse
	early_stopping_rounds	50
		stages without a better validation log likelihood
	use_best_iteration	True
		predictions use the best stage, not every stage fitted
	ngboost_ensemble_seeds	1
		1 = one fit, no seed ensemble
	n_estimators	300
	max_depth	None
		no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
<i>Kernel models</i>		

Configuration	Hyperparameter	Default value
SVM svm	C	1.0
	gamma kernel	scale rbf
GP gauche_rbf	kernel	rbf
		set by this job, not by the shared spec
	likelihood_noise	0.001
		starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000
		molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1
		used only when the standard fitter refuses the model
	fallback_adam_iters	100
		used only when the standard fitter refuses the model
GP-Tanimoto gauche run on ecfp4 only	single_thread_fit	False
		a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True
		the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500
		molecules sampled to estimate that median
	collapse_fraction	0.05
		a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
	kernel	tanimoto
		set by this job, not by the shared spec
	likelihood_noise	0.001
		starting value; it is learned
	outputscale apply_outputscale	1.0 True

Configuration	Hyperparameter	Default value
GP-Hetero heteroscedastic_gp	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cu- bic in this. The test set is un- touched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fit- ter refuses the model
	fallback_adam_iters	100 used only when the standard fit- ter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library de- fault
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the train- ing label spread is recorded as collapsed rather than scored
	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cu- bic in this. The test set is un- touched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fit- ter refuses the model
	fallback_adam_iters	100 used only when the standard fit- ter refuses the model

Configuration	Hyperparameter	Default value
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
	noise network hidden sizes	[64, 64] models/models.py, not in the shared spec
	noise network output	Softplus, plus 10^{-4} models/models.py, not in the shared spec
	Adam learning rate, process	0.1 models/models.py, not in the shared spec
	Adam learning rate, noise network	0.001 models/models.py, not in the shared spec
	noise network epochs	100 the shared epoch cap
<i>Neural networks</i>		
NN- α dnn	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
NN- β mlp	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
BNN- α dnn_bnn_full	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules

Configuration	Hyperparameter	Default value
BNN- β mlp_bnn_full	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
VBLL- α dnn_bnn_full_variational	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL- β mlp_bnn_full_variational	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
VBLL- α (noise head) dnn_bnn_full_variational_hetero	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL- β (noise head) mlp_bnn_full_variational_hetero	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above

Configuration	Hyperparameter	Default value
BNN- α (variance head) dnn_bnn_full_mve	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo
	output head	1 / number of training molecules value and log variance two outputs
	loss	Gaussian negative log likeli- hood
	log variance clamp	[-10, 10] scripts/loss_functions.py, not in the shared spec
BNN- β (variance head) mlp_bnn_full_mve	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo
	output head	1 / number of training molecules value and log variance two outputs
	loss	Gaussian negative log likeli- hood
	log variance clamp	[-10, 10] scripts/loss_functions.py, not in the shared spec

Additional file 1, continued

Additional file 1, Table B: the tuned settings that replace a default, and the four datasets they apply to. A setting is chosen once per model per dataset, not per representation, and is then used for all six representations: a two-way comparison across model and representation cannot be read if each cell carries its own hyperparameters. Only the Bayesian and variational networks are tuned; every other configuration in Table A trains at the default. A model with no row for a dataset kept the default on that dataset. Learning rates are printed to three significant figures; the source files hold the full value. Source files: `results/master_tuned_hyperparameters.json` for QM9 and `results/master_tuned_hyperparameters.lab.json` for the three assay datasets.

Dataset	Configuration	Hyperparameter	Tuned value
QM9	BNN- α	activation	tanh
		hidden_size1	64
		hidden_size2	32
QM9	VBLL- α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	BNN- β	dropout_rate	0.379
		hidden_size	64
		lr	0.00429
		num_hidden_layers	1
QM9	VBLL- β	dropout_rate	0.357
		hidden_size	64
		lr	0.00119
		num_hidden_layers	1
Caco-2	VBLL- α	activation	relu
		hidden_size1	64
		hidden_size2	32
Caco-2	BNN- β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
hERG K_i	VBLL- α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	BNN- β	dropout_rate	0.169
		hidden_size	64
		lr	0.00602
		num_hidden_layers	2
LogD	VBLL- α	activation	tanh
		hidden_size1	256
		hidden_size2	32
LogD	VBLL- β	dropout_rate	0.079
		hidden_size	128
		lr	0.00011
		num_hidden_layers	2

Additional file 2

Table S2: Pairwise Spearman rank correlations between model noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap ($\rho \geq 0.95$). “Model A” and “Model B” label each pair, both treated symmetrically. Checkmarks indicate pairs where one model was excluded from the ANOVA due to near-redundancy. Gauche (RBF) pairs omitted (all $\rho = 1.0$, $n = 6$, PDV-only artifact).

Model A	Model B	n	ρ	Excluded
VBLL- β	NGBoost	18	0.998	
LightGBM	XGBoost	46	0.998	
Gauche	VBLL- β	18	0.996	✓
QRF	RF	40	0.995	✓
VBLL- α	LightGBM	24	0.992	
RF	XGBoost	46	0.989	
NGBoost	XGBoost	40	0.988	
NGBoost	RF	40	0.987	
LightGBM	RF	46	0.986	
VBLL- β	SVM	24	0.986	
QRF	XGBoost	40	0.986	✓
NGBoost	QRF	34	0.986	✓
VBLL- α	NGBoost	18	0.986	
VBLL- α	XGBoost	24	0.984	
LightGBM	NGBoost	40	0.984	
LightGBM	QRF	40	0.984	✓
VBLL- α	VBLL- β	24	0.983	
NN- α	NN- β	52	0.983	
LightGBM	SVM	46	0.982	
BNN- β	SVM	34	0.981	
VBLL- α	SVM	24	0.979	
BNN- α	VBLL- β	24	0.979	
VBLL- α	QRF	24	0.978	✓
SVM	XGBoost	46	0.978	
BNN- α	BNN- β	36	0.977	
BNN- α	SVM	34	0.976	
VBLL- α	Gauche	18	0.975	✓
VBLL- α	RF	24	0.975	
QRF	SVM	40	0.974	✓
BNN- α	VBLL- α	24	0.974	
LightGBM	VBLL- β	24	0.973	
VBLL- α	BNN- β	24	0.972	
Gauche	SVM	40	0.972	✓
Gauche	XGBoost	44	0.970	✓
NGBoost	SVM	40	0.970	
NN- α	BNN- β	36	0.970	
Gauche	LightGBM	42	0.969	✓
RF	SVM	46	0.969	
BNN- β	VBLL- β	24	0.968	
VBLL- β	QRF	24	0.966	✓
NN- α	BNN- α	36	0.965	
VBLL- β	XGBoost	24	0.965	
NN- β	SVM	46	0.962	
NN- α	SVM	46	0.960	
Gauche	NGBoost	34	0.959	✓
LightGBM	BNN- β	36	0.959	
BNN- α	NN- β	36	0.958	
BNN- α	QRF	28	0.956	✓
Gauche	QRF	34	0.956	✓
VBLL- β	RF	24	0.955	
Gauche	RF	40	0.954	✓
BNN- β	XGBoost	34	0.953	
BNN- β	QRF	28	0.952	✓
NN- α	VBLL- β	24	0.952	
NN- β	BNN- β	36	0.952	

Model A	Model B	n	ρ	Excluded
BNN- α	LightGBM	36	0.951	
NN- β	VBLL- β	24	0.950	
BNN- α	Gauche	30	0.950	✓

Additional file 3

Table S3: Pairwise Spearman rank correlations between representation noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap. “Rep A” and “Rep B” label each pair of molecular representations, both treated symmetrically. Checkmarks indicate pairs where one representation was excluded from the ANOVA.

Rep. A	Rep. B	n	ρ	Excluded
ECFP4	Morgan	60	0.995	✓
MHGGNN	Morgan	48	0.994	✓
Rand. SMILES	SMILES	16	0.994	✓
Rand. SMILES	SNS	16	0.994	✓
ECFP4	MHGGNN	54	0.993	
ECFP4	SNS	56	0.992	✓
Morgan	SNS	52	0.992	✓
MHGGNN	SNS	44	0.991	✓
PDV	ECFP4	72	0.990	
PDV	MHGGNN	48	0.989	
SMILES	SNS	52	0.988	✓
Morgan	Rand. SMILES	16	0.988	✓
Morgan	SMILES	54	0.988	✓
PDV	Morgan	54	0.986	✓
PDV	Rand. SMILES	12	0.986	✓
PDV	SNS	50	0.984	✓
ECFP4	SMILES	72	0.984	
Mol2Vec	Morgan	48	0.982	✓
MHGGNN	Mol2Vec	54	0.982	
ECFP4	Mol2Vec	54	0.982	
PDV	PDV (binary)	72	0.980	✓
MHGGNN	SMILES	48	0.980	
PDV	Mol2Vec	48	0.979	
ECFP4	Rand. SMILES	16	0.979	✓
PDV	SMILES	66	0.978	
MHGGNN	PDV (binary)	54	0.972	✓
Mol2Vec	SNS	44	0.969	✓
Mol2Vec	SMILES	48	0.966	
ECFP4	PDV (binary)	78	0.965	✓
Morgan	PDV (binary)	60	0.962	✓
PDV (binary)	SNS	56	0.961	✓
Mol2Vec	PDV (binary)	54	0.956	✓
MHGGNN	Rand. SMILES	16	0.956	✓
PDV (binary)	SMILES	72	0.941	✓
PDV (binary)	Rand. SMILES	16	0.929	✓
Mol2Vec	Rand. SMILES	16	0.891	✓

Additional file 4

Table S4: Intraclass correlation coefficients (ICC(1,1)) for model pairs with $ICC \geq 0.5$, assessing within-family consistency of noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap. ICC(1,1) measures the proportion of total variance attributable to between-configuration differences; values near 1.0 indicate that two models rank configurations almost identically. Lower (more negative) NDS indicates faster degradation under noise. The Family column groups each pair: “Tree” (both tree-based), “ α/β ” (architecture-pair partners such as BNN- α /BNN- β), “ α ” (both share the α neural-network backbone), or “Cross” (cross-family pair).

Model A	Model B	n	ICC(1,1)	Mean $ \Delta NDS $	Family
LightGBM	XGBoost	8	0.944	0.0060	Tree
BNN- α	BNN- β	6	0.846	0.0101	α/β
QRF	RF	7	0.810	0.0216	Tree
VBLL- α	VBLL- β	4	0.782	0.0108	α/β
QRF	XGBoost	7	0.754	0.0157	Tree
LightGBM	QRF	7	0.743	0.0133	Tree
VBLL- α	LightGBM	4	0.698	0.0115	Cross
VBLL- β	SVM	4	0.689	0.0133	Cross
RF	XGBoost	8	0.678	0.0212	Tree
VBLL- α	XGBoost	4	0.656	0.0133	Cross
BNN- α	VBLL- α	4	0.604	0.0134	α
LightGBM	RF	8	0.586	0.0244	Tree
LightGBM	VBLL- β	4	0.577	0.0174	Cross
LightGBM	SVM	8	0.561	0.0117	Cross
BNN- α	VBLL- β	4	0.552	0.0123	α/β
VBLL- β	XGBoost	4	0.544	0.0195	Cross
VBLL- α	QRF	4	0.540	0.0229	Cross
VBLL- α	SVM	4	0.522	0.0124	Cross
Gauche	VBLL- β	3	0.519	0.0113	Cross

Additional file 5

Table S5: Configurations excluded from robustness analysis due to baseline $R^2 \leq 0.6$ on the QM9 HOMO–LUMO gap, summarized by model and representation. Each cell shows the number of noise strategies (out of 6) for which the configuration was excluded.

Model	Mol2Vec	MHGGNN	SMILES	Rand. SMILES
BNN- α	6	6	—	—
BNN- β	6	6	—	—
VBLL- α	6	6	—	—
VBLL- β	6	6	—	—
RF	—	—	—	4
QRF	—	—	—	4
NGBoost	—	—	6	4

Additional file 6

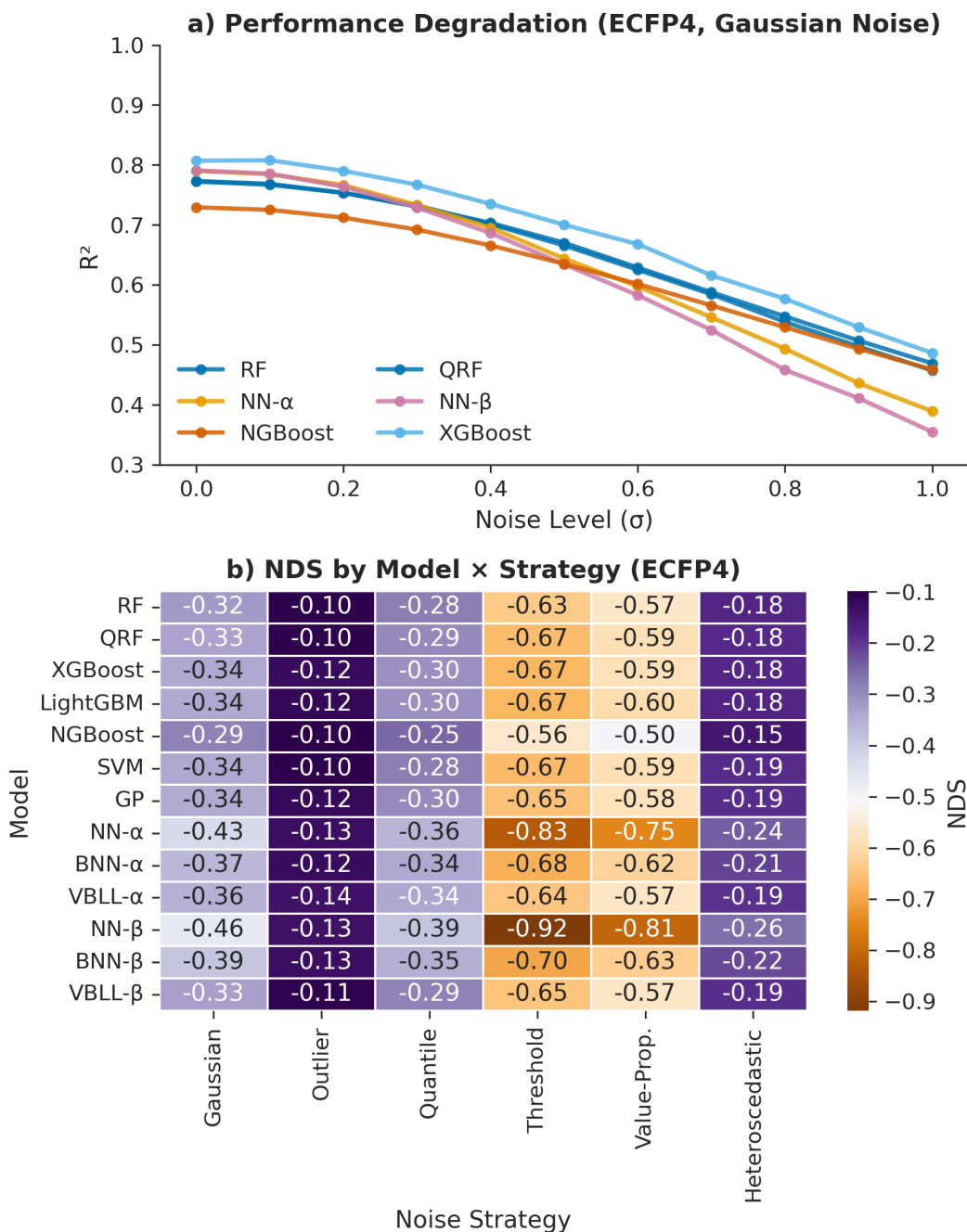


Figure S1: Global overview of noise robustness on the ECFP4 representation, evaluated on QM9 HOMO–LUMO gap. a) Performance degradation curves (R^2 versus noise level σ) under Gaussian noise. b) Noise degradation slope (NDS) heatmap across model architectures and noise strategies; lower (more negative) NDS indicates faster degradation under noise.

Additional file 7

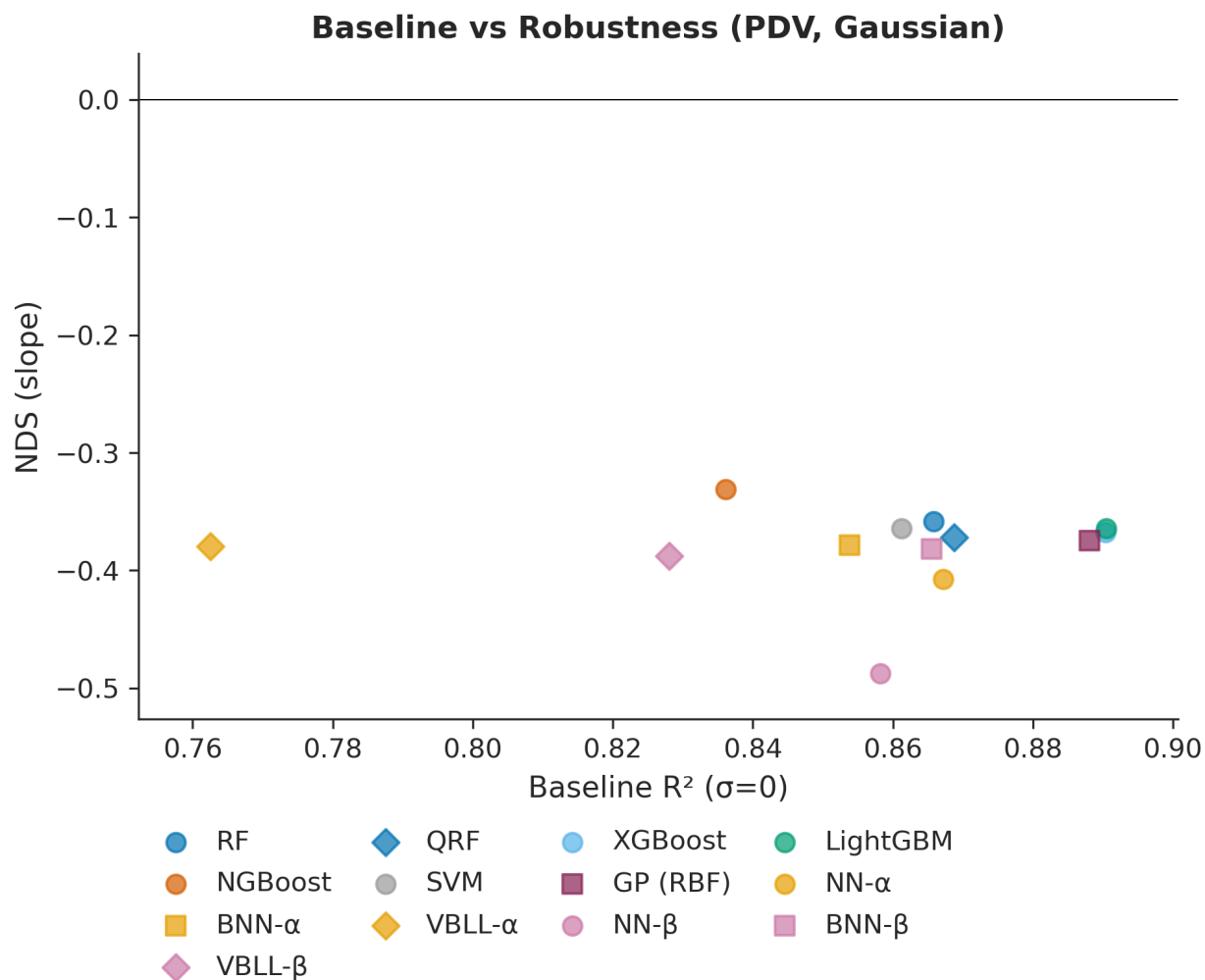


Figure S2: Baseline R^2 versus noise degradation slope (NDS) scatter for all models on the physicochemical descriptor vector (PDV) representation under Gaussian noise, on the QM9 HOMO–LUMO gap. Lower (more negative) NDS indicates faster performance loss under noise.

Additional file 8

Table S6: Neural network Bayesian transformation effects on noise degradation slope (NDS) by representation, on the QM9 HOMO–LUMO gap. Positive Δ NDS indicates the Bayesian variant is more robust (less negative slope) than its deterministic counterpart.

Model	Representation	Baseline R^2	NN- β NDS	BNN- β NDS	Δ NDS
NN- β	ECFP4	0.791	−0.464	−0.388	+0.076
NN- β	PDV	0.858	−0.487	−0.382	+0.106
NN- β	SMILES	0.721	−0.558	−0.365	+0.193

Additional file 9

Table S7: Uncertainty quantification metrics for probabilistic models on the ECFP4 representation (Gaussian noise strategy) on the QM9 HOMO–LUMO gap. Unc-Error ρ : Spearman correlation between predicted uncertainty and absolute prediction error. Unc-Noise ρ : Spearman correlation between predicted uncertainty and injected noise level σ . ECE: expected calibration error (lower = better-calibrated). Coverage targets: 68% (1σ), 95% (2σ).

Model	Unc-Error ρ	Unc-Noise ρ	ECE	Cov. 1σ	Cov. 2σ
QRF	0.25	0.27	0.19	74.7%	94.6%
NGBoost	0.22	0.37	0.11	69.2%	94.2%
GP	0.18	0.34	0.23	78.6%	96.0%
BNN- β	0.20	0.37	0.29	78.6%	95.2%
BNN- α	0.15	0.32	0.24	76.6%	95.1%
VBLL- β	0.19	0.34	0.19	73.9%	95.5%
VBLL- α	0.11	0.31	0.22	74.5%	95.1%

Additional file 10

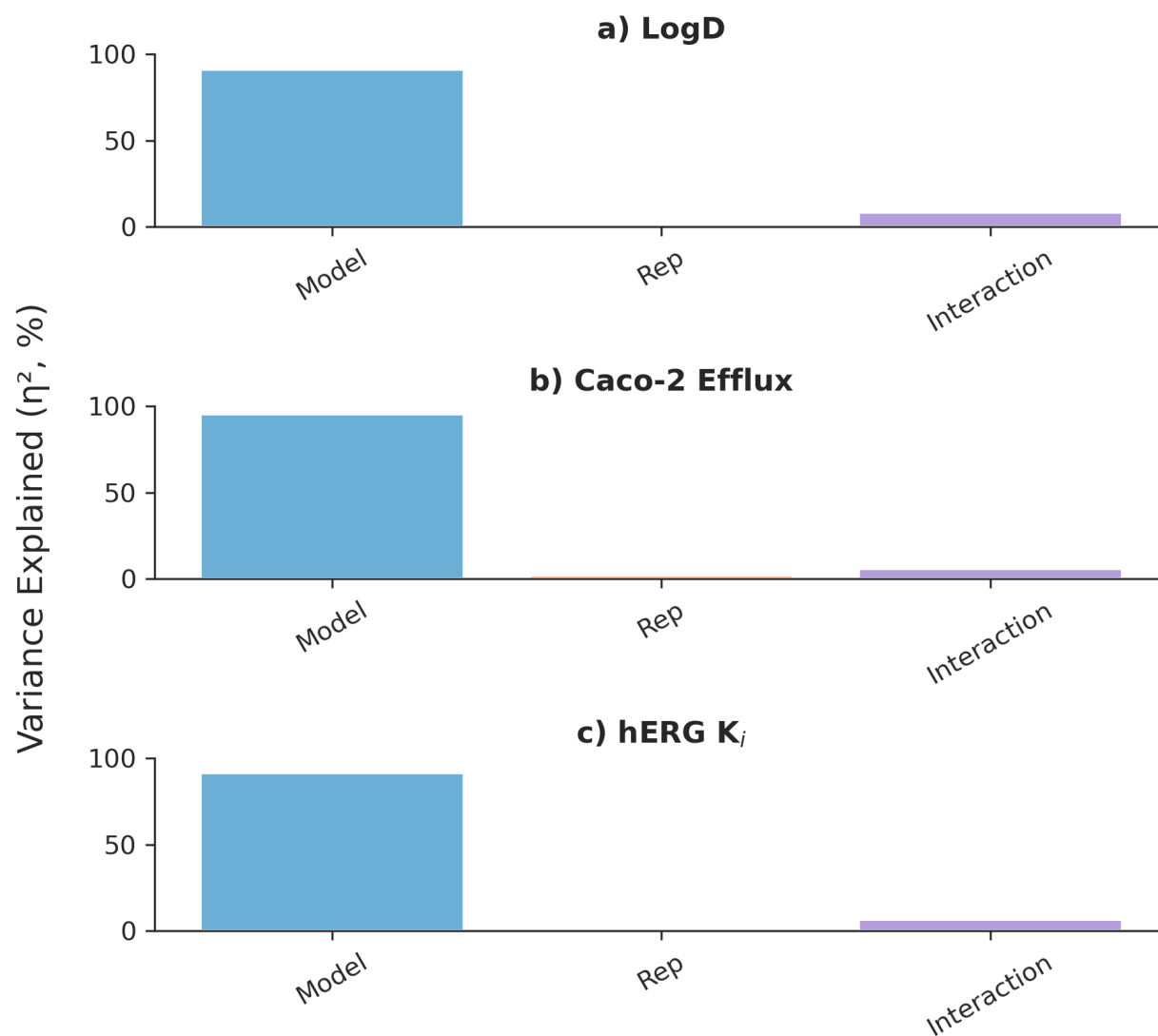


Figure S3: ANOVA variance decomposition (η^2 , %) on external validation datasets: a) LogD, b) Caco-2 Efflux, and c) hERG K_i . Model, representation, and interaction contributions to noise degradation slope (NDS) variance for each dataset; lower (more negative) NDS indicates faster performance loss under noise.

Additional file 11

Table S8: Random forest (RF) vs quantile regression forest (QRF) noise degradation slope (NDS) on the three external validation datasets (LogD, Caco-2 Efflux, hERG K_i). Lower (more negative) NDS indicates faster degradation under noise.

Dataset	RF NDS	QRF NDS	Δ NDS
ChEMBL-hERG-Ki	-0.231	-0.359	-0.128
OpenADMET-Caco2	-0.534	-0.627	-0.094
OpenADMET-LogD	-0.101	-0.185	-0.083